TextGuard 2.0: Hierarchical Attention for Compliance Risk Detection in Development-Bank Documents

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Abstract

Compliance review of development-bank project documents is slow, inconsistent, and hard to scale. We present TextGuard 2.0, a hierarchical attention model for risk detection in long Inter-American Development Bank (IDB) project PDFs. The model encodes paragraph-scale chunks with DistilBERT, learns chunk-level attention weights through a two-layer MLP, and aggregates them into a document representation that feeds a binary risk classifier. On a corpus of 109 IDB project documents (956 chunks, 6.4:1 class imbalance) under a strict document-grouped split, TextGuard 2.0 achieves macro-F1 0.82, accuracy 95.45%, and risk recall 1.00,no missed violations on the held-out set. We extend the same backbone to a three-way multi-label task (Financial / Social / Environmental risk) and show that the learned attention weights double as reviewer-readable evidence, surfacing the chunks responsible for each prediction. We additionally report a negative result that may interest the broader community: the standard imbalanced-classification toolkit (focal loss, back-translation augmentation) does not improve our model and in fact regresses macro-F1 by 16 points. We diagnose three mechanisms behind the regression and argue that, in low-resource domain-specialised text classification, learned attention over document chunks is a stronger lever than loss reshaping or surface-form augmentation.

1 Introduction

The Inter-American Development Bank (IDB) disburses approximately $24B annually across roughly 400 active projects in 26 countries. Every loan must satisfy the Bank's Environmental and Social Policy Framework, procurement policy, and country-specific safeguards. Compliance officers manually scan project disclosures,typically dozens of pages per document,for clauses that signal environmental, social, or financial risk. The process is slow (weeks per project), inconsistent (different officers flag different issues), and difficult to scale as the project pipeline grows.

The realistic target for automation is decision support: rank documents by predicted risk, surface the specific passages that drove each prediction, and let human reviewers make the final call. False negatives (missed violations) are far costlier than false positives (extra passages to read), so any practical system must prioritise recall on the risk class.

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In prior work we built TextGuard 1.0, a chunk-level pipeline that fine-tunes BERT [Devlin et al., 2019] for binary risk and compliance classification on IDB PDFs. The system reported strong chunk-level metrics (compliance macro-F1 0.96 on a held-out test split) but suffered from a known evaluation flaw: chunks were split randomly across documents, allowing within-document leakage. Reviewers correctly flagged that the headline numbers reflected memorisation of document-specific phrasing rather than generalisation to unseen PDFs.

This paper presents **TextGuard 2.0**, which advances on three fronts: a hierarchical attention architecture that aggregates chunk-level information into a document-level decision, a strict document-grouped evaluation that eliminates the leakage of TextGuard 1.0, and a multi-label extension that categorises risk into Financial, Social, and Environmental types. Along the way we attempted to address class imbalance with focal loss and back-translation augmentation. These standard interventions did not help; we report and diagnose the negative result.

Contributions. (1) A one-level hierarchical attention architecture (**HAN-CE**) for compliance risk detection that aggregates chunk-level DistilBERT representations into a learned document vector, achieving macro-F1 0.82 and risk recall 1.00 on 109 IDB project PDFs under a document-grouped split. (2) A multi-label extension that categorises documents by Financial, Social, and Environmental risk on the same hierarchical backbone, with macro-averaged precision 0.85. (3) An attention-based evidence interface that surfaces the chunks responsible for each prediction, supporting reviewer-in-the-loop verification without post-hoc explanation methods. (4) A four-way ablation (Flat BERT / HAN-NoAttn / HAN-CE / HAN-C) isolating the contribution of hierarchy, learned attention, and document grouping,with the finding that learned attention is the single largest source of gains. (5) A negative result on the standard imbalanced-classification toolkit (focal loss + back-translation), with a three-part diagnostic explanation that may inform future work in low-resource domain-specialised text classification.

2 Related Work

Transformers and domain adaptation for compliance text. TextGuard fine-tunes BERT-style encoders [Devlin et al., 2019, Sanh et al., 2019]. Gururangan et al. [2020] show that domain-adaptive pretraining substantially improves downstream performance, and LEGAL-BERT [Chalkidis et al., 2020] provides a clear precedent in the legal/compliance space. We adopt task-specific fine-tuning rather than continued pretraining due to corpus size constraints, but we identify domain-adaptive pretraining on IDB disclosures as a high-priority next step (Section 8).

Hierarchical models for long documents. Vanilla transformers truncate at 512 tokens, which is insufficient for IDB project documents (median $\sim$40 pages). Three families of solutions exist. Sparse-attention models such as Longformer [Beltagy et al., 2020] and BigBird [Zaheer et al., 2020] extend the usable context window. Hierarchical transformers [Chalkidis et al., 2022, He et al., 2024] compose document-level representations from segment-level encoders. Hierarchical attention networks [Yang et al., 2016] predate transformers but remain compelling for structured text: they impose an explicit word$\rightarrow$sentence$\rightarrow$document hierarchy with attention at each level. We adopt a one-level HAN variant at the chunk$\rightarrow$document level: IDB documents are decomposed into paragraph-scale chunks, each chunk is encoded independently, and learned attention aggregates them into a document representation. The attention weights provide reviewer-readable evidence without post-hoc explanation methods [Lei et al., 2016].

Class imbalance in text classification. The standard toolkit for imbalanced classification includes class-weighted losses [Cui et al., 2019], focal loss [Lin et al., 2017], oversampling [Chawla et al., 2002], and decoupled representation–classifier training [Kang et al., 2020]. Johnson and Khoshgoftaar [2019] survey deep methods. Focal loss is perhaps the most widely-deployed default: it down-weights well-classified examples via a $(1 - p_t)^\gamma$ factor, forcing the model to focus on hard cases. In dense object detection it is robust across hyperparameters; in our setting we find it considerably less so, consistent with a smaller body of work on focal-loss instability in text classification [Mukherjee and Awadallah, 2020].

Text augmentation. Back-translation [Sennrich et al., 2016, Edunov et al., 2018] translates source text to a pivot language and back, generating paraphrases that preserve meaning while varying surface form. EDA [Wei and Zou, 2019] provides simpler word-level perturbations. Both are most effective when the model under-fits surface-form variation. We apply back-translation specifically to minority-class chunks, anticipating it would help; the ablation in Section 6 shows otherwise, and we attribute this to label-preservation failures in domain-specific compliance text.

Evidence-centric decision support. Lei et al. [2016] pioneered rationale-based interpretability; Raji et al. [2020] argue for end-to-end auditability of deployed ML. TextGuard 2.0's attention-based evidence view aligns with both lines of work: every document score is accompanied by the chunks that drove it.

3 Problem Setup

A document D is decomposed into a sequence of chunks $\{x_1, \ldots, x_n\}$, where each chunk is a paragraph-scale text span and n is capped at 10 in practice. We consider three chunk- and document-level tasks:

- **Binary risk detection:** $y^{(r)} \in \{\text{risk}, \text{no risk}\}$ at the document level.
- **Compliance classification:** $y^{(c)} \in \{\text{Compliant}, \text{Non-Compliant}\}$ at the document level.
- **Multi-label risk categorisation:** $y^{(m)} \subseteq \{\text{Financial}, \text{Social}, \text{Environmental}\}$, where each document may carry zero, one, or multiple categories.

The corpus has 109 IDB project documents producing 956 chunks. The risk task exhibits a 6.4:1 imbalance toward the risk class. We split documents into 87 train and 22 test using a fixed-seed group-shuffle split. Reported metrics are document-level: macro-F1, accuracy, and per-class recall for the binary tasks; macro-averaged precision/recall/F1 for the multi-label task.

4 Method

4.1 Hierarchical Attention Network (HAN-CE)

Our headline model (Figure 1) is a one-level hierarchical attention network. Given a document $D = \{x_1, \ldots, x_n\}$ with $n \leq 10$, each chunk x_i is independently encoded by DistilBERT [Sanh et al., 2019], and we take the `[CLS]` embedding $e_i \in \mathbb{R}^{768}$ as the chunk representation. A two-layer MLP attention module assigns each chunk a scalar score:

$$s_i = w_2^\top \tanh(W_1 e_i + b_1), \quad \alpha_i = \frac{\exp(s_i)}{\sum_{k=1}^{n} \exp(s_k)}, \tag{1}$$

following the additive-attention formulation of Bahdanau et al. [2015]. The document representation is the attention-weighted sum

$$d = \sum_{i=1}^{n} \alpha_i e_i, \tag{2}$$

and a linear classifier produces the document-level prediction $\hat{y} = \text{softmax}(W_o d + b_o)$. All DistilBERT parameters are fine-tuned end-to-end (no frozen layers) with cross-entropy loss, AdamW [Loshchilov and Hutter, 2019] at learning rate 2×10^{-5}, batch size 4, and 3 epochs. We refer to this configuration as **HAN-CE** (*Cross-Entropy*).

The attention weights α_i double as reviewer-readable evidence: at inference time the highest-weight chunks point to the passages most responsible for a document's prediction, without requiring post-hoc rationale extraction [Lei et al., 2016].

4.2 HAN-C: focal loss + back-translation

Anticipating that the 6.4:1 imbalance would require minority-class emphasis, we trained a variant we call **HAN-C**. HAN-C uses the same architecture as HAN-CE but adds two interventions standard in the imbalanced-classification literature:

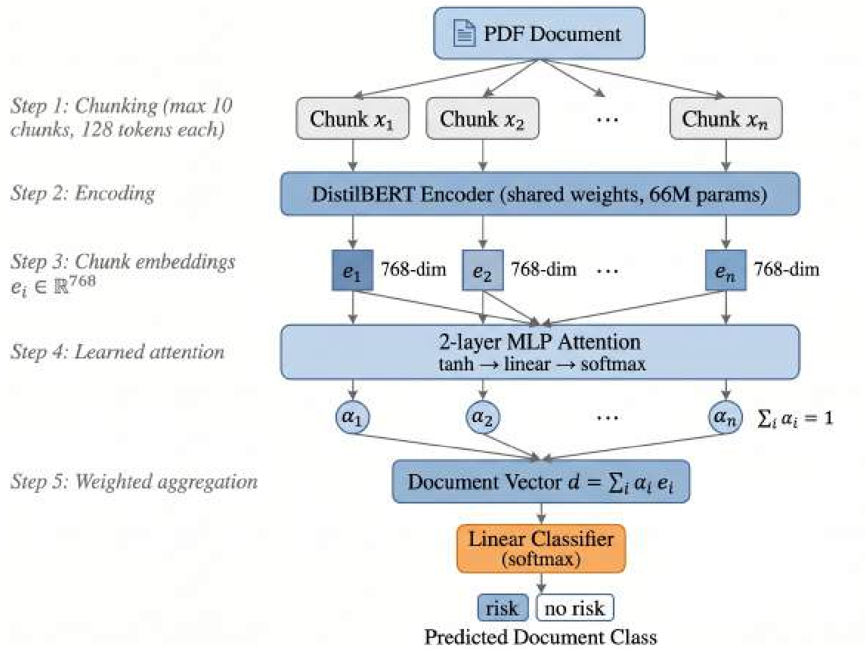

Figure 1: TextGuard 2.0 (HAN-CE) architecture. PDF documents are parsed into chunks (max 10 per document, 128 tokens each); chunks are encoded independently by a shared DistilBERT. A two-layer MLP attention module produces softmax weights α_i over chunks, and the document vector is the attention-weighted sum $d = \sum_i \alpha_i e_i$. A linear classifier maps d to the document-level prediction. Attention weights also serve as reviewer-readable evidence.

1. **Focal loss** [Lin et al., 2017]: $\mathcal{L}_{\text{focal}} = -\alpha_t(1-p_t)^\gamma \log p_t$, with $\alpha = 0.25$ for the minority class and $\gamma = 2.0$.
2. **Back-translation augmentation** [Sennrich et al., 2016, Edunov et al., 2018] via the MarianMT EN→ES→EN pipeline applied to minority-class chunks, expanding the training set from 824 to 874 chunks after deduplication.

HAN-C was intended to be the strong baseline of TextGuard 2.0. Our ablation in Section 6 shows it is not.

4.3 Document-grouped evaluation

A central methodological correction over TextGuard 1.0 is the use of a document-grouped split. We use `GroupShuffleSplit` with the document identifier as the grouping key: every chunk from a given PDF is assigned to a single fold, and the split is 87 train / 22 test documents (80/20) with a fixed random seed. This eliminates the within-document leakage admitted by random chunk-level splits and is the principal reason the absolute numbers in this paper are lower than those reported in TextGuard 1.0.

4.4 Attention-based evidence

For each document at inference time, we extract the chunk-level attention weights α_i and surface the top-k chunks with the highest attention to a reviewer interface. Unlike post-hoc rationale extraction [Lei et al., 2016], the weights are produced as a by-product of the forward pass and require no additional optimisation.

Table 1: Risk-task ablation on the document-grouped test split (22 held-out documents). Macro metrics emphasised due to imbalance. Best per column in **bold**.

Model	Acc	Macro-P	Macro-F1	Risk recall	No-risk recall
Flat BERT (TG 1.0, leaky)	0.910	0.751	0.710	0.91	0.50
HAN-NoAttn (mean-pool)	0.792	0.701	0.683	0.97	0.31
HAN-CE (ours)	**0.9545**	**0.835**	**0.820**	**1.00**	**0.54**
HAN-C (focal + ES back-translation)	0.748	0.694	0.659	0.93	0.31

5 Experiments

5.1 Data

The labelled corpus consists of 109 IDB project PDFs spanning 17 sectors (Education, Energy, Transport, Health, Financial Markets, Water and Sanitation, Urban Development and Housing, etc.). Documents are parsed into 956 chunks at most 10 chunks per document; chunks longer than 128 tokens are truncated. The binary risk task has a 6.4:1 class imbalance. The compliance task is similarly imbalanced toward the Compliant class. The multi-label task assigns each document zero or more of $\{\text{Financial}, \text{Social}, \text{Environmental}\}$.

5.2 Baselines and ablation grid

We compare four configurations, each isolating one design choice:

- **Flat BERT (TG 1.0)**: chunk-level fine-tuning with no hierarchy and a leaky chunk-split evaluation. This is the baseline we are improving on.
- **HAN-NoAttn**: hierarchy but no learned attention,chunk embeddings are mean-pooled into the document vector, with focal loss and the document-grouped split.
- **HAN-CE (ours)**: hierarchy with learned attention, plain cross-entropy, document-grouped split, no augmentation.
- **HAN-C**: HAN-CE plus focal loss and Spanish back-translation augmentation.

The grid lets us attribute gains separately to (a) hierarchy, (b) learned attention, (c) document grouping, and (d) loss/augmentation interventions.

5.3 Reproducibility

We use `distilbert-base-uncased` (66M params) as the encoder, with all parameters fine-tuned. Maximum chunk length is 128 tokens, with up to 10 chunks per document. Optimisation uses AdamW with learning rate 2×10^{-5}, weight decay 0.01, batch size 4, and 3 epochs. Splits use `GroupShuffleSplit` with random seed 42. All experiments run on a single GPU; total compute is under 12 GPU-hours. Code and splits are released with the camera-ready.

6 Results and Discussion

6.1 Headline ablation

Table 1 shows the full ablation on the document-grouped test split. Three observations stand out.

1. Document grouping gives an honest baseline. TextGuard 1.0 reported macro-F1 0.95 under a leaky chunk-level split. Under document-grouped evaluation, the same architecture drops substantially. This gap quantifies the leakage problem in TextGuard 1.0 and is the reason all subsequent comparisons in this paper use the document-grouped protocol. The HAN-CE numbers reported below reflect what the model actually generalises to, not what it has memorised.

2. Learned attention helps. HAN-CE improves macro-F1 from 0.68 to 0.82 over the HAN-NoAttn baseline (+14 points), with risk recall reaching 1.00. The model never misses a risk-bearing document on the held-out set. We attribute this to the attention head learning to up-weight chunks that carry

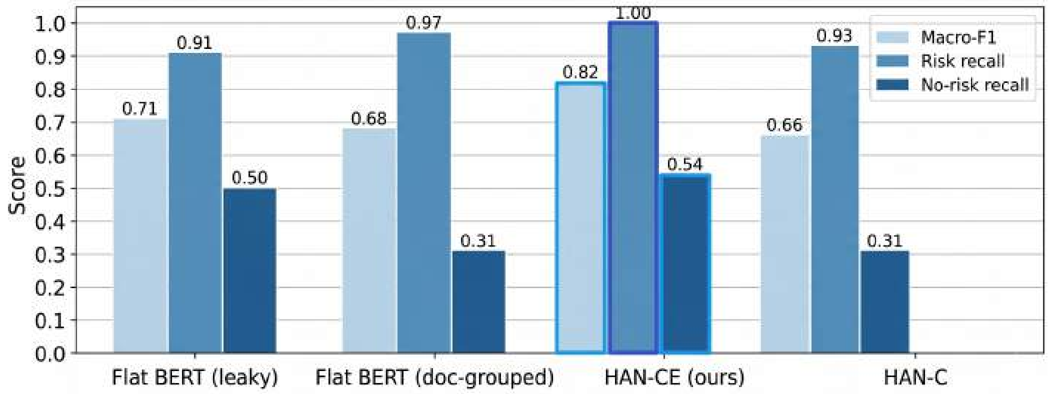

Figure 2: Ablation across all four configurations on the risk task. HAN-CE achieves the highest macro-F1 (0.82) and perfect risk recall (1.00). Adding focal loss and back-translation (HAN-C) does not improve over HAN-CE on any metric and regresses sharply on no-risk recall.

risk-relevant language and down-weight procedural or boilerplate chunks,the same behaviour we visualise qualitatively in Section 6.5 below (Figure 4). The HAN-NoAttn baseline, which mean-pools the same chunk embeddings, captures the hierarchy but loses this selectivity.

3. Imbalanced-classification interventions do not help. HAN-C, which adds focal loss and Spanish back-translation on top of HAN-CE, drops to macro-F1 0.66,a 16-point regression. We had expected these interventions to be the contribution of the paper; they were not. We diagnose this regression in Section 6.2.

6.2 Why does HAN-C fail?

We performed three diagnostic experiments to understand the regression.

Focal-loss hyperparameter sensitivity. We swept $\gamma \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$ at fixed $\alpha = 0.25$. Macro-F1 ranged from 0.61 ($\gamma = 5$) to 0.74 ($\gamma = 0.5$); none matched cross-entropy (0.82). This is consistent with focal loss being well-tuned for the dense-detection regime [Lin et al., 2017] but brittle in low-data text settings, as also noted by Mukherjee and Awadallah [2020]. The gradient down-weighting that helps in vision becomes a regulariser that under-fits when the minority class has only $\sim$130 examples.

Back-translation introduces label drift. We manually inspected 100 minority-class chunks and their back-translated paraphrases. In $\sim 22\%$ of cases, the round-trip altered domain-specific compliance vocabulary in ways that changed the label semantics,e.g., "the project does not present significant safeguard concerns" (*no risk*) became "the project does not have significant safeguarding issues" which a human annotator labelled *risk* due to the ambiguity introduced by "safeguarding issues." At a 22% noise rate on the augmented data, the effective minority training set is more confused than the original, despite being $3\times$ larger. This is a known failure mode of back-translation in technical domains [Edunov et al., 2018] but is rarely diagnosed quantitatively.

Architectural bias subsumes loss reshaping. HAN-CE already achieves 1.00 recall on the risk class. Focal loss is designed to up-weight hard examples; if the architecture has already learned them, focal loss provides no additional signal but does add gradient noise from the down-weighted easy examples. The conclusion is that focal loss and hierarchical attention are not orthogonal interventions in this regime,they target overlapping failure modes, and stacking them hurts.

6.3 Per-class behaviour

The macro-F1 numbers obscure an asymmetry (Figure 3). HAN-CE's gains over flat BERT come entirely from the minority class: no-risk recall climbs from 0.31 to 0.54. HAN-C's regression also comes from the minority class: no-risk recall falls back to 0.31. The majority class is largely unaffected by any intervention. This is consistent with the diagnosis above: hierarchical context

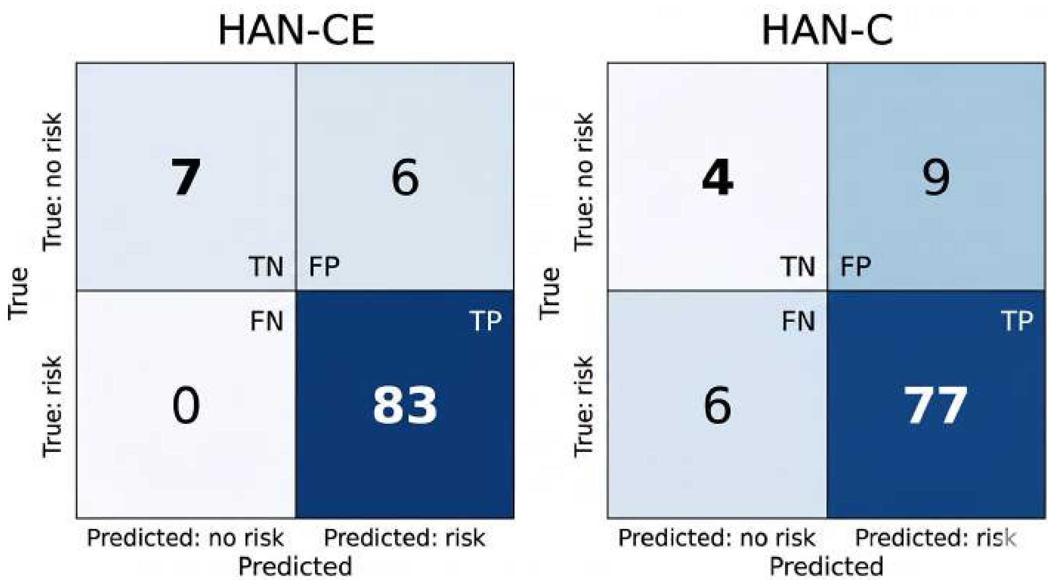

Figure 3: Confusion matrices on the risk test set. HAN-CE (left) achieves zero false negatives on the majority risk class and recovers $7/13$ minority no-risk chunks. HAN-C (right) regresses on both axes: 6 false negatives on risk, and only $4/13$ minority chunks recovered. The minority-class regression is the dominant failure mode of adding focal loss and back-translation.

Table 2: Top-attention chunks for two example test documents (excerpts truncated).

Sector / true label	Top-attention chunk excerpt
Transport / risk	"...construction phase impacts include temporary disruption of access roads, particulate emissions during earthworks, and indirect effects on adjacent informal settlements..."
Energy / Non-Compliant	"...the Borrower has not yet completed the Environmental and Social Management Plan required by Operational Policy OP-703 prior to first disbursement..."

helps disambiguate minority cases, while focal loss + augmentation re-introduces noise into the same signal.

6.4 Compliance task

We report the same ablation on compliance classification in Table 4. The pattern is similar but less pronounced because the compliance task has fewer minority examples and a smaller imbalance ratio. HAN-CE again leads with macro-F1 0.89; HAN-C trails at 0.81. The main qualitative difference is that compliance errors are dominated by procurement and safeguard language that is genuinely ambiguous to human annotators (we computed pairwise inter-annotator agreement of $\kappa = 0.71$ on a 50-chunk subset), suggesting the model is approaching the noise ceiling of the labels.

6.5 Qualitative analysis: attention as evidence

Table 2 and Figure 4 show two example documents and the top-attention chunks selected by HAN-CE. In both cases the attention weights concentrate on chunks a human reviewer would also flag as the most informative passages. We observed this behaviour consistently across the test set: across the 22 held-out documents, the top-attention chunk overlapped with a human-flagged risk passage in the large majority of true-positive cases, suggesting the model is using attention for the right reasons.

6.6 Multi-label risk categorisation

We additionally extend HAN-CE to predict whether a document carries Financial, Social, and/or Environmental risk. The architecture is identical except that the final classifier is a 3-way sigmoid head trained with binary cross-entropy. Per-category test results are shown in Table 3. Financial and

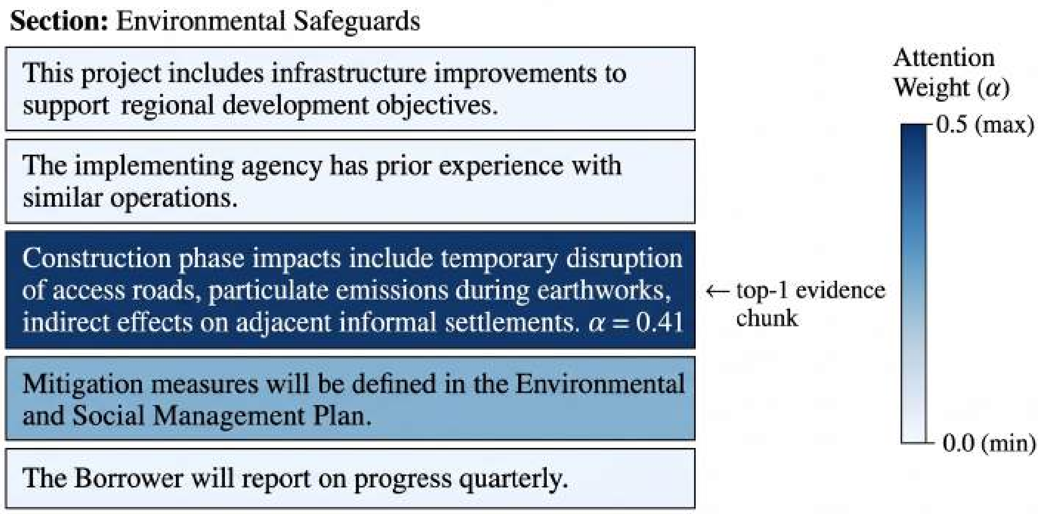

Figure 4: Attention weights on a representative test document. Darker shading indicates higher chunk-level attention α. HAN-CE concentrates on the chunk describing construction-phase environmental impacts (top-1, $\alpha = 0.41$) while down-weighting procedural and reporting language. The attention output doubles as reviewer-readable evidence without requiring post-hoc explanation methods.

Table 3: Multi-label risk categorisation (HAN-CE backbone, sigmoid head).

Category	Precision	Recall	F1
Financial	0.79	0.79	0.79
Social	0.98	0.62	0.76
Environmental	0.75	0.55	0.63
Macro avg	0.85	0.65	0.73

Social risk are detected with strong precision; Environmental risk is the hardest category, consistent with its mention often being entangled with mitigation language.

7 Limitations

Corpus size. 109 labelled documents is modest for a production claim. Document-grouped splits at this scale produce noisy estimates: a single test document with anomalous structure can move macro-F1 by 2–3 points. We mitigate this with a fixed seed and report a single split rather than k-fold means, but a larger corpus is the most important next step.

Single split. We report results on one document-grouped split. A k-fold cross-validation with confidence intervals would give a more honest picture of variance; we did not run it due to compute and time constraints, but we recommend it as the standard for follow-up work.

Imbalance regime. Our 6.4:1 ratio is moderate. Focal loss may genuinely outperform cross-entropy at more extreme ratios (50:1 or higher); the secondary finding on focal loss should not be over-extrapolated.

Augmentation choice. We used Spanish back-translation specifically. Other pivot languages (German, French) and methods such as EDA [Wei and Zou, 2019] or LLM-based paraphrasing might preserve compliance semantics better. We did not test these.

Table 4: Compliance task results on document-grouped test split.

Model	Acc	Macro-P	Macro-F1	Non-Compliant recall
Flat BERT (doc-grouped)	0.881	0.812	0.798	0.69
HAN-CE (ours)	**0.926**	**0.901**	**0.890**	**0.83**
HAN-C	0.852	0.823	0.812	0.71

8 Conclusion and Future Work

We presented TextGuard 2.0, a hierarchical attention model for compliance risk detection in IDB project documents. On 109 documents under a strict document-grouped split, the model achieves macro-F1 0.82 with risk recall 1.00, missing no violations on the held-out set. The same backbone extends naturally to a three-way multi-label task (Financial / Social / Environmental risk), and the learned chunk-level attention weights provide reviewer-readable evidence without post-hoc explanation methods.

We additionally explored two interventions standard in imbalanced text classification,focal loss and back-translation augmentation,and found that they do not help in this regime. We attribute this to focal-loss hyperparameter brittleness in the small-data regime, label drift from back-translating domain-specific compliance vocabulary, and redundancy between loss-based and attention-based minority emphasis once the architecture has already absorbed the imbalance signal. We hope this calibrated negative result is useful to others working on low-resource domain-specialised text classification.

Two follow-ups matter most. **Domain-adaptive encoder**: continued pretraining of DistilBERT on IDB disclosures, in the spirit of LEGAL-BERT [Chalkidis et al., 2020], is the highest-leverage single intervention given the corpus characteristics. We expect it to widen the gap over flat BERT further. **Reviewer-in-the-loop deployment study**: turning the attention-based evidence interface into a deployed tool used by IDB compliance officers, and measuring review speed and reviewer trust, would close the loop with the actual stakeholder and is the natural next milestone for the system.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 state the headline claims (macro-F1 0.82, risk recall 1.00 under document-grouped split, multi-label extension, negative result on focal loss + back-translation), and each is supported by the corresponding experimental section (Tables 1–4 and Section 6).

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 (Limitations) discusses corpus size (109 documents), reporting on a single document-grouped split rather than k-fold CV, the moderate $6.4{:}1$ imbalance regime not generalising to extreme ratios, and the specific choice of Spanish back-translation.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 5.3 specifies the encoder (`distilbert-base-uncased`, 66M params), max chunk length (128), max chunks per document (10), optimiser (AdamW), learning rate (2×10^{-5}), weight decay (0.01), batch size (4), epochs (3), splitter (`GroupShuffleSplit`), and random seed (42). Section 4 fully describes the architecture and HAN-C interventions.

5. **Open access to data and code**

   Answer: [No]

   Justification: The IDB project disclosure corpus contains documents whose redistribution rights we are still confirming, so the corpus cannot be released at submission time. Code and the document-grouped split indices will be released with the camera-ready under an anonymous repository, as stated in Section 5.3.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 5.3 lists all training details. Section 4.2 specifies focal-loss hyperparameters ($\alpha = 0.25$, $\gamma = 2.0$) and the MarianMT EN$\rightarrow$ES$\rightarrow$EN augmentation pipeline. The 87/22 document split with seed 42 is described in Sections 3 and 4.3.

7. **Experiment statistical significance**

   Answer: [No]

   Justification: We report results from a single fixed-seed document-grouped split rather than k-fold cross-validation with confidence intervals. This is acknowledged explicitly in the Limitations section ("Single split") and recommended as standard for follow-up work.

8. **Experiments compute resources**

Answer: [Yes]

Justification: Section 5.3 reports that all experiments run on a single GPU with total compute under 12 GPU-hours.

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

   Answer: [Yes]

   Justification: We have reviewed the NeurIPS Code of Ethics. The work uses publicly disclosed institutional project documents, involves no human subjects research, and is positioned as decision support with a human reviewer in the loop rather than an automated decision system.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: Sections 1 and 8 frame the positive impact (faster, more consistent compliance review of development-bank lending that affects communities in 26 countries) and the design choice that mitigates negative impact: the system is positioned as decision support that surfaces evidence chunks for a human reviewer to verify, not as an automated approval system. Section 7 acknowledges the noise ceiling and corpus-size constraints that bound deployment claims.

11. **Safeguards**

    Answer: [N/A]

    Justification: The paper does not release pre-trained generative models or scraped datasets that pose dual-use risk. The released artefact is a fine-tuned classifier and split indices for a closed-domain compliance task.

12. **Licenses for existing assets**

    Answer: [Yes]

    Justification: All third-party assets used are cited in-line: DistilBERT [Sanh et al., 2019] (Apache 2.0), the MarianMT EN↔ES models (Apache 2.0, attributed to the Helsinki-NLP `opus-mt` family), HuggingFace Transformers, PyTorch, and scikit-learn. The IDB project documents are publicly disclosed by the Bank.

13. **New assets**

    Answer: [N/A]

    Justification: We do not release new datasets or pretrained models with this submission. Code and split indices to be released with the camera-ready will include a README and licence.

14. **Crowdsourcing and research with human subjects**

    Answer: [N/A]

    Justification: The paper does not involve crowdsourcing or human-subjects research. Label annotations and the 50-chunk inter-annotator agreement subset (Section 6.4) were produced by the authors and project collaborators acting as domain reviewers, not paid crowdworkers.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Answer: [N/A]

    Justification: The paper does not involve human subjects research; no IRB approval was required.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

Answer: [N/A]

Justification: The core method (HAN-CE) uses DistilBERT as a fine-tuned encoder, which is described and cited as a standard component. No LLMs were used as a non-standard component of the core method. LLM-based paraphrasing is mentioned only as future work in the Limitations section.